

De novo Biotin Biosynthesis via the BioC–BioH Route: A Review

1. Executive Summary

De novo biotin biosynthesis in bacteria is best understood as a two-stage biosynthetic system. In the first ("front-end") stage, the seven-carbon α,ω -dicarboxylic acid pimelate moiety is manufactured; in the second ("back-end") stage, the pimelate thioester is elaborated into the fused bicyclic ring system of the finished cofactor. The defining innovation of the **BioC–BioH route**, the subject of this review, is how the pimelate moiety is made: rather than dedicating a bespoke set of enzymes to build a C7 diacid, the pathway **hijacks the host's type II fatty acid synthase (FAS II)**. BioC, a SAM-dependent O-methyltransferase, methyl-esterifies the free ω -carboxyl of malonyl-ACP, disguising this atypical, charged substrate so that the standard FAS II elongation enzymes accept it. Two rounds of elongation extend the masked primer to pimeloyl-ACP methyl ester, and the esterase **BioH** then hydrolyzes the methyl ester to give pimeloyl-ACP plus methanol, unmasking the intermediate and halting further elongation at C7 (Lin, Hanson & Cronan 2010, [PMID: 20693992](#); Lin & Cronan 2012, [PMID: 22965231](#)).

The pathway thus resolves into a **plastic front end** and a **conserved back end**. The methylation/demethylation logic is a recurring theme, but the specific enzyme performing the demethylation step is remarkably interchangeable across lineages: BioH is frequently replaced by non-orthologous esterases (BioG, BioK, BioV, Tam-associated activities) or bypassed entirely by alternative first-stage chemistries (BioZ; the BioW/BioI route). By contrast, the ring-assembly enzymes — **BioF** (8-amino-7-oxononanoate/KAPA synthase), **BioA** (diaminononanoate aminotransferase), **BioD** (dethiobiotin synthetase), and **BioB** (radical-SAM biotin synthase) — are strongly conserved and operate in an obligate linear order. The pathway culminates in one of biochemistry's most striking reactions: BioB inserts a sulfur atom between two unactivated carbons of dethiobiotin, sacrificing its own auxiliary [2Fe-2S] cluster as the sulfur donor and thereby behaving essentially as a single-turnover enzyme in vitro.

This review defines the boundaries of the biosynthetic module (distinguishing it from biotin transport and BirA-mediated regulation, which are neighboring but separate systems), lays out the best current mechanistic model, catalogues the major molecular players, surveys

evolutionary and cell-biological variation (including the eukaryotic BioA–BioD gene fusion with substrate channeling), and identifies the constraints, controversies, and open questions that remain. Eight findings, anchored to primary literature, form the evidentiary backbone.

2. Definition and Biological Boundaries

What is included. The BioC–BioH biosynthetic system, as modeled here, comprises six pathway-specific enzymatic reactions carried out by six enzyme families:

1. **BioC** — malonyl-ACP O-methyltransferase (masks the ω -carboxyl).
2. **BioH** — pimeloyl-ACP methyl ester esterase (unmasks it; the C7 gatekeeper).
3. **BioF** — 8-amino-7-oxononanoate (KAPA/AON) synthase.
4. **BioA** — adenosylmethionine-8-amino-7-oxononanoate transaminase (DAPA aminotransferase).
5. **BioD** — dethiobiotin synthetase.
6. **BioB** — biotin synthase.

What is an explicit dependency but not a member. Between BioC and BioH, the generic FAS II elongation machinery — the condensing enzymes (FabB/FabF), ketoreductase (FabG), dehydratase (FabA/FabZ), and enoyl reductase (FabI) acting on the ACP-tethered, methyl-esterified intermediate — performs two elongation cycles. This machinery is shared with membrane lipid synthesis and is not biotin-specific; it is a dependency of the system rather than a member of it. Its precursor is malonyl-ACP, itself derived from acetyl-CoA carboxylase (ACC) activity (reviewed in [PMID: 34132100](#)).

What lies outside the boundary. Two neighboring processes are routinely conflated with biosynthesis but should be treated separately:

- **Biotin transport/uptake.** BioY-type S-components, sometimes assembled into energy-coupling-factor (ECF) BioMNY transporters, import biotin from the environment ([PMID: 25991724](#)). Transport is an alternative to synthesis, not a step within it.
- **BirA-mediated regulation and biotin attachment.** The bifunctional BirA both ligates biotin to acceptor proteins (biotin protein ligase activity) and, in many bacteria, represses the biotin operon transcriptionally in response to biotinoyl-AMP levels ([PMID: 30808984](#); Cronan 2024, [PMID: 37997490](#)). This regulatory/attachment layer bounds the pathway (it controls how much enzyme is made and consumes the finished product) but is

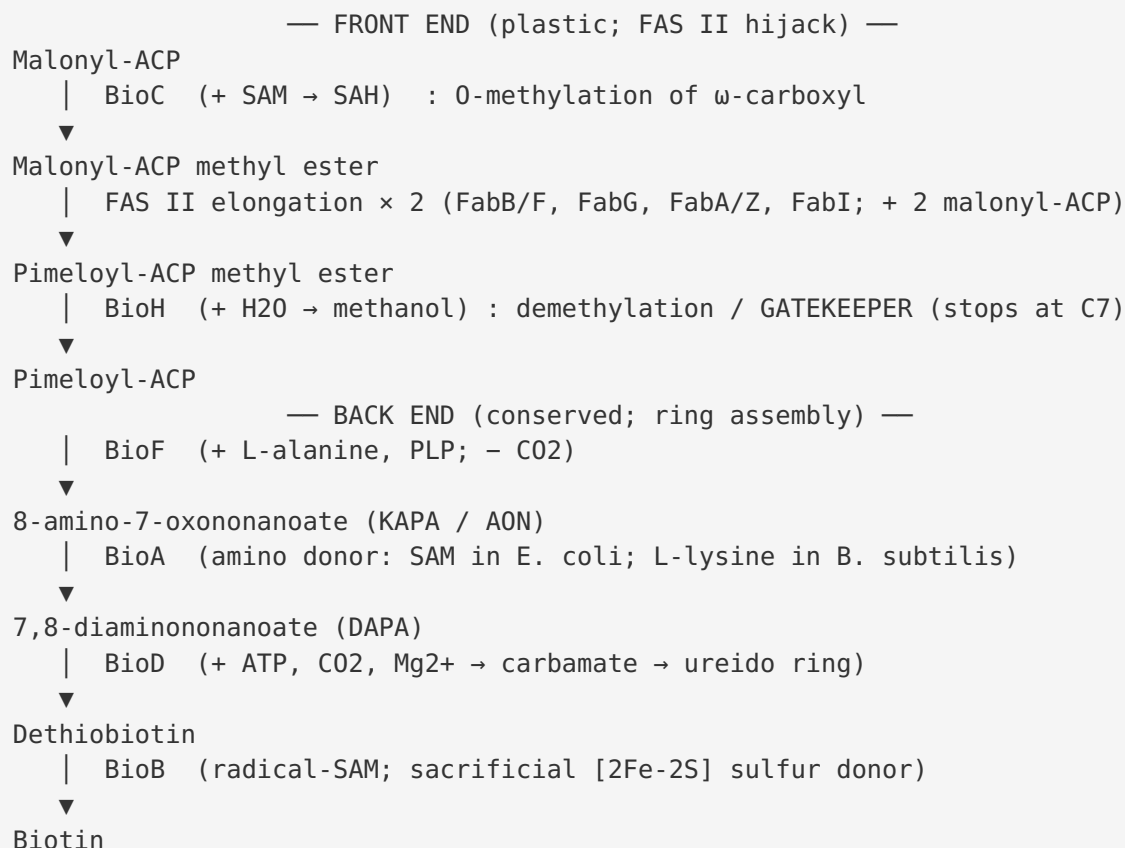
mechanistically distinct from the six biosynthetic reactions. Alternative repressors exist — BioR in α -proteobacteria ([PMID: 26037461](#)) and XRE-type BioX ([PMID: 33972354](#)) — and lineage-specific DNA-binding mechanisms differ (E. coli BirA requires biotinoyl-AMP-induced dimerization; S. aureus BirA binds DNA as a monomer, [PMID: 30808984](#)).

A subtle boundary case is **dethiobiotin salvage**: some bacteria carrying a *bioY-bioB* gene cluster import dethiobiotin and convert it to biotin using only BioB ([PMID: 37640259](#)). This engages the pathway's terminal enzyme alone and blurs the synthesis/salvage boundary at the last step.

Competing definitions. The literature sometimes uses "biotin biosynthesis" to mean the entire homeostatic network (synthesis + transport + BirA regulation). For mechanistic clarity this review adopts the narrower six-enzyme definition and treats the front end (pimelate synthesis) and back end (ring assembly) as functionally distinct sub-modules — a division strongly supported by their contrasting evolutionary behavior.

3. Mechanistic Overview

The best current model is an ordered, two-stage linear pathway. The steps and their obligatory ordering are as follows:



Obligatory steps. Every arrow above is obligatory in the canonical route, and the ordering is enforced by substrate chemistry: each enzyme acts on the product of the previous one. The back-end sequence BioF → BioA → BioD → BioB is a strict linear cascade with no known shortcuts (dethiobiotin salvage excepted, which enters at the last step).

Conditional / variable steps. The **demethylation** step (BioH) is obligatory in function but conditional in identity — different lineages use different enzymes (Section 5). The **entire front end** is conditional at the level of strategy: organisms using free pimelic acid (*B. subtilis*) or glutaryl-CoA priming (Rhizobiales BioZ) do not use BioC-then-FAS-then-BioH at all.

Accessory / dependency components. FAS II elongation is accessory to biotin synthesis in the sense that it is borrowed, not dedicated. ACC (malonyl-CoA/ACP supply) and the Fe-S cluster biogenesis machinery (which reloads BioB's sacrificial cluster) are upstream dependencies without which the pathway cannot run but which are not biotin-specific.

Molecular assemblies carrying out each step. BioC is a Rossmann-fold SAM-dependent methyltransferase. BioH is an α/β -hydrolase-fold serine esterase (Ser-His-Asp triad). BioF is a homodimeric, PLP-dependent α -oxoamine synthase. BioA is a fold-type-I PLP aminotransferase. BioD is an ATP/Mg²⁺-dependent carbamoyltransferase/synthetase. BioB is a homodimeric radical-SAM enzyme carrying two distinct iron-sulfur clusters per monomer.

4. Major Molecular Players and Active Assemblies

4.1 BioC — malonyl-ACP O-methyltransferase (Step 1)

BioC introduces the "disguise." Direct enzymatic characterization of *Bacillus cereus* BioC (which complements *E. coli* in vivo) established that **malonyl-ACP is a far better methyl acceptor than malonyl-CoA**, that BioC is specific for the malonyl moiety, and that it is competitively inhibited by S-adenosyl-L-homocysteine and sinefungin — hallmarks of a bona fide SAM-dependent O-methyltransferase (Lin & Cronan 2012, [PMID: 22965231](#)). A key physiological consequence is **toxicity on overexpression**: high-level BioC expression blocks growth and fatty-acid synthesis, because unchecked methylation drains malonyl-ACP away from membrane lipid production. This explains why the pathway must throttle BioC tightly. *E. coli* BioC is notoriously difficult to purify in active form, so much of the mechanistic understanding comes from non-*E. coli* orthologs — a caveat for cross-organism generalization. BioC has recently been framed as a druggable anti-ESKAPE target (Su et al. 2024, [PMID: 39705367](#)).

4.2 BioH — pimeloyl-ACP methyl ester esterase and gatekeeper (Step 2)

BioH removes the disguise and, crucially, **defines the chain length** of the product. A 2.05-Å co-crystal structure of *E. coli* BioH bound to pimeloyl-ACP methyl ester demonstrated that **BioH — not BioF — is the gatekeeper**, and that its physiological substrate is pimeloyl-ACP methyl ester (Agarwal et al. 2012, [PMID: 23045647](#)). This resolved a genuine ambiguity: in isolation BioH is a nonspecific carboxylesterase that will cleave many substrates, and BioF could in principle also perform the gatekeeping function. The structure showed that chain-length specificity is enforced not by the active site alone but by **protein–protein recognition of the ACP-bound substrate**. Mechanistically BioH uses a canonical **Ser-His-Asp catalytic triad** typical of lipolytic α/β -hydrolases (e.g., a metagenomic BioH with the Ser78-His230-Asp202 triad, [PMID: 23413993](#)).

4.3 BioF — 8-amino-7-oxononanoate (KAPA) synthase (Step 3)

BioF is a PLP-dependent α -oxoamine synthase that condenses the pimeloyl-thioester with **L-alanine**, releasing CO₂ and forming KAPA (8-amino-7-oxononanoate). It initiates the committed ring-assembly stage. Structural work on *Mycobacterium smegmatis* BioF confirmed an active-site organization similar to *E. coli* AONS and identified PLP-binding residues (His129, Lys235, His200) essential for activity in vivo ([PMID: 25462832](#)).

4.4 BioA — DAPA aminotransferase (Step 4)

BioA transaminates KAPA to 7,8-diaminononanoate (DAPA). Notably, the **amino donor varies by lineage**: *E. coli* BioA uses **S-adenosylmethionine (SAM)** as the amino donor, whereas *Bacillus subtilis* BioA uses **L-lysine** (Van Arsdell et al. 2005, [PMID: 15880481](#)). This is a well-documented example of mechanistic variation within a conserved step and a caution against assuming the SAM-donor mechanism is universal.

4.5 BioD — dethiobiotin synthetase (Step 5)

BioD closes the ureido (first) ring of the bicyclic system. It is an **ATP/Mg²⁺-dependent** enzyme that carboxylates the 7-amino group of DAPA to a carbamate, activates it as a mixed carbamic-phosphoric anhydride, and cyclizes to dethiobiotin. *Mycobacterium tuberculosis* DTBS is **nucleotide-promiscuous** — it utilizes ATP, CTP, GTP, ITP, TTP, or UTP with similar K_m and k_{cat} — owing to the loss of a gatekeeper Asn present in *E. coli* DTBS (Salaemae et al. 2015, [PMID: 25801336](#); [PMID: 30289406](#)). This makes BioD an actively pursued anti-TB drug target.

4.6 BioB — biotin synthase (Step 6)

BioB catalyzes the pathway's terminal and most chemically demanding reaction: **oxidative insertion of a sulfur atom between the C6 methylene and the C9 methyl of dethiobiotin** to form biotin's thiophane ring (Cramer & Jarrett 2018, [PMID: 30097099](#)). BioB is a radical-SAM enzyme carrying two clusters per monomer:

- A **[4Fe-4S] cluster** that binds SAM and reductively cleaves it to a 5'-deoxyadenosyl radical (the C–H-abstracting species). Spectroscopy shows this cluster is stable during turnover and is present in the SAM-bound form (Jameson et al. 2004, [PMID: 14967042](#); [PMID: 12440894](#)).
- An **auxiliary [2Fe-2S] cluster** that is the **direct sulfur donor** and is destroyed during catalysis.

The sacrificial nature of the auxiliary cluster is the defining — and controversial — feature of BioB (Section 7).

Molecular players at a glance:

Step	Enzyme	Fold / class	Cofactors / cosubstrates	Product
1	BioC	SAM-dependent O-MTase	SAM → SAH	Malonyl-ACP methyl ester
(dep.)	FAS II	Various	Malonyl-ACP, NADPH	Pimeloyl-ACP methyl ester
2	BioH	α/β -hydrolase esterase (Ser-His-Asp)	H ₂ O → methanol	Pimeloyl-ACP
3	BioF	PLP α -oxoamine synthase	PLP, L-alanine; – CO ₂	KAPA (AON)
4	BioA	PLP aminotransferase	SAM (E. coli) / L-lysine (B. subtilis)	DAPA
5	BioD	ATP/Mg ²⁺ synthetase	ATP (or other NTP), CO ₂	Dethiobiotin
6	BioB	Radical-SAM	[4Fe-4S]+SAM, sacrificial [2Fe-2S]	Biotin

5. Evolutionary and Cell-Biological Variation

The system's evolutionary signal is bimodal: **the front end is plastic; the back end is conserved.**

5.1 Plasticity of the pimelate-synthesis (front-end) stage

The *bioH* gene is **absent from many *bioC*-containing genomes** and is functionally replaced by evolutionarily distinct esterases ([PMID: 26868423](#); Shapiro, Chakravartty & Cronan 2012, [PMID: 23152908](#)):

Front-end variant	Organisms	Mechanism / role
BioH	<i>E. coli</i> and many Gram-negatives	Canonical pimeloyl-ACP methyl ester esterase
BioG, BioK	Various bacteria	Non-orthologous esterases; complement Δ bioH and cleave the methyl ester in vitro (PMID: 23152908)
BioV	<i>Helicobacter</i>	Lineage-restricted esterase replacing BioH (PMID: 26868423)
BioH1/2/3 + Tam/BioC	<i>Mycobacterium smegmatis</i>	Three isoenzymes; defines a "BioC-BioH(3)" paradigm (PMID: 35816546)
BioZ (bypass)	Rhizobiales	Initiates via glutaryl-CoA condensation; bypasses BioC-BioH (PMID: 33824341)
BioW/BioI (bypass)	<i>Bacillus subtilis</i>	BioI (cytochrome P450) cleaves fatty acyl chains to free pimelate; BioW (pimeloyl-CoA synthetase) activates free pimelic acid (PMID: 28196402 ; PMID: 21437340)

The take-home is that **producing a pimeloyl-thioester is the invariant requirement**, but multiple non-homologous molecular solutions (convergent at the level of chemistry) have arisen to reach it. As the authors of [PMID: 23152908](#) put it, BioH, BioG and BioK "catalyze a common reaction" yet "the proteins are evolutionarily distinct."

5.2 Conservation of the ring-assembly (back-end) stage

BioF, BioA, BioD, and BioB are found together across the biotin-synthesizing bacterial and eukaryotic world with strong sequence conservation, reflecting the difficulty and specificity of the chemistry (PLP condensation, transamination, ATP-dependent cyclization, radical sulfur insertion). Their conservation, contrasted with front-end plasticity, supports treating the pathway as two evolutionarily independent sub-modules.

5.3 Eukaryotic elaboration: the BioA–BioD gene fusion with channeling

In plants and most fungi, the DAPA aminotransferase (BioA) and dethiobiotin synthetase (BioD) activities are fused into a **single bifunctional polypeptide** encoded by a gene formed from the fusion of prokaryotic monofunctional ancestors. In *Arabidopsis*, the **BIO3-BIO1 fusion protein** is the sole product of the locus, is targeted to mitochondria, and **channels the DAPA**

intermediate directly between the two active sites (Cobessi et al. 2012, [PMID: 22547782](#)). This is a clear later elaboration — a lineage-specific architectural innovation (physical linkage + substrate channeling + organellar targeting) layered on top of the conserved catalytic core. Consistent with a mitochondrial locale for the back end, BioB and lipoyl synthase both draw on mitochondrial Fe-S cluster supply, and their sacrificial chemistry imposes a demand on that supply chain in plants ([PMID: 36846908](#)).

5.4 Lineage-specific tuning of individual back-end enzymes

Even within the conserved back end, individual enzymes are tuned per lineage: BioA's amino donor (SAM vs. L-lysine; [PMID: 15880481](#)) and BioD's nucleotide selectivity (strict ATP vs. promiscuous NTP use; [PMID: 25801336](#)) are the best-documented examples. Recent bioinformatic surveys even reveal variation in BioB's auxiliary-cluster coordination, including a 4Fe-5S cluster in some homologs (Lachowicz et al. 2024, [PMID: 38215281](#)).

6. Constraints, Dependencies, and Failure Modes

6.1 Ordering constraints (what must happen in what order)

The known precedence relationships are strict and chemically enforced:

1. Malonyl-ACP → malonyl-ACP methyl ester (BioC) **before** demethylation (BioH).
2. Demethylation (BioH) **before** KAPA formation (BioF).
3. KAPA formation (BioF) **before** DAPA formation (BioA).
4. DAPA formation (BioA) **before** dethiobiotin ring closure (BioD).
5. Dethiobiotin ring closure (BioD) **before** biotin formation (BioB).

The methyl ester **must** be installed before FAS elongation (it is what allows FAS to accept the substrate) and **must** be removed after elongation (pimeloyl-ACP, not its methyl ester, is BioF's usable precursor). This mutual dependence — mask first, elongate, then unmask — is the logical core of the route and rules out any path in which FAS acts on the free (unmasked) diacid. This is the key piece of evidence that excludes an otherwise-plausible "direct dicarboxylate elongation" model: the charged ω -carboxyl is incompatible with FAS acceptance unless masked ([PMID: 20693992](#)).

6.2 Compartment and substrate constraints

- **ACP tethering.** Front-end intermediates are ACP-bound; recognition of the ACP (not just the small-molecule moiety) governs both FAS acceptance and BioH gatekeeping ([PMID: 23045647](#)). This is a substrate-presentation constraint.
- **Substrate specificity of BioC.** Malonyl-ACP (not malonyl-CoA) is the physiological acceptor ([PMID: 22965231](#)); this couples the pathway obligately to the ACP pool and hence to FAS II.
- **Chain length.** The system stops at C7 (pimelate). Gatekeeping by BioH — enforced through protein recognition rather than an inability of FAS to elongate further — is what prevents overshoot to C9 and beyond.
- **Compartmentalization in eukaryotes.** The back end is mitochondrion-targeted in plants (the BIO3-BIO1 fusion; [PMID: 22547782](#)), tying flux to mitochondrial Fe-S and one-carbon metabolism.

6.3 Failure modes and throughput bottlenecks

- **BioC overexpression toxicity.** Because BioC diverts malonyl-ACP from membrane lipid synthesis, excess BioC activity halts growth and fatty-acid synthesis ([PMID: 22965231](#)). The pathway must be dosed carefully — a built-in failure mode that also explains the need for BirA repression of the operon.
 - **BioB single-turnover bottleneck.** Because BioB consumes its own [2Fe-2S] cluster as the sulfur source, it produces only **~0.7–1 equivalent of biotin per monomer in vitro** ([PMID: 14967042](#); [PMID: 30097099](#)). Sustained biotin production therefore requires **continuous Fe-S cluster regeneration**, making BioB the pathway's key throughput constraint and coupling biotin output to cellular iron-sulfur biogenesis capacity.
 - **Loss of the front end.** Many host-associated and intracellular bacteria lack a complete de novo pathway and depend on transport (BioY/BioMN) — a physiological "failure mode" resolved by scavenging rather than synthesis ([PMID: 25991724](#); [PMID: 37640259](#)).
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7. Controversies and Open Questions

7.1 The BioB sulfur-donor mechanism

The strongest evidence that BioB's auxiliary **[2Fe-2S] cluster is the direct sulfur donor** comes from selenium-substitution experiments: reconstituting BioB with a [2Fe-2Se] cluster and assaying yielded **selenobiotin**, directly tracing the inserted chalcogen to the auxiliary cluster rather than to free sulfide (Tse Sum Bui et al. 2006, [PMID: 16533066](#)). Spectroscopy shows the [2Fe-2S] cluster is degraded during turnover while the [4Fe-4S]/SAM cluster remains stable ([PMID: 14967042](#)). This "sacrificial cluster" model is now widely accepted ([PMID: 38215281](#)).

However, the mechanism remains debated in detail. The observation that the [2Fe-2S] cluster decays about an order of magnitude faster than biotin forms indicates that, **if** the cluster is the immediate donor, sulfur insertion is not rate-limiting — or, alternatively, that the cluster is **not** the immediate donor and instead generates a protein-bound persulfide/polysulfide intermediate that is the true proximal sulfur source ([PMID: 14967042](#)). Chalcogenide exchange complicates interpretation of the selenium experiments ([PMID: 16533066](#)). And the discovery of unexpected variation in auxiliary-cluster coordination — including a **4Fe-5S** cluster in some BioBs ([PMID: 38215281](#)) — shows the architecture is not monolithic. The precise identity of the immediate sulfur species and the in vivo cluster-regeneration mechanism remain the central open questions for the terminal step.

7.2 Gatekeeper identity, historically

Assignment of BioH's physiological substrate was long "problematical" because BioH is a nonspecific esterase and BioF could plausibly perform the same gatekeeping cut ([PMID: 23045647](#)). The co-crystal structure settled this in favor of BioH, but the general lesson — that an enzyme's in vitro promiscuity can obscure its physiological role — recurs across the front end and cautions against inferring function from in vitro activity alone.

7.3 Organism-mixing and generalization hazards

Much mechanistic detail is stitched together across organisms: BioC biochemistry from *B. cereus* (because *E. coli* BioC resists purification; [PMID: 22965231](#)); BioA donor chemistry differing between *E. coli* and *B. subtilis* ([PMID: 15880481](#)); BioD nucleotide behavior differing between *E. coli* and *M. tuberculosis* ([PMID: 25801336](#)); and the eukaryotic fusion architecture from *Arabidopsis* ([PMID: 22547782](#)). These are legitimately different systems, and the field's

habit of assembling a "consensus pathway" from them risks over-generalizing lineage-specific features. The front-end plasticity documented above makes clear that no single organism's route should be presented as *the* biotin pathway.

7.4 Salvage/synthesis boundary

The finding that *bioY–bioB* organisms import dethiobiotin and finish it with BioB alone ([PMID: 37640259](#)) raises the question of how common such hybrid synthesis/salvage strategies are, and whether the terminal step is best considered part of "biosynthesis" or "salvage" in these lineages.

8. Key References

Core pathway definition and front-end mechanism - Lin, Hanson & Cronan 2010 — *Biotin synthesis begins by hijacking the fatty acid synthetic pathway*. [PMID: 20693992](#). Establishes the BioC methylation / FAS elongation / BioH demethylation model (Finding F001). - Lin & Cronan 2012 — *The BioC O-methyltransferase catalyzes methyl esterification of malonyl-ACP..* [PMID: 22965231](#). Malonyl-ACP specificity, SAM dependence, overexpression toxicity (F005). - Agarwal et al. 2012 — *Structure of the enzyme-ACP substrate gatekeeper complex...* [PMID: 23045647](#). Identifies BioH as the C7 gatekeeper (F002). - Shapiro, Chakravartty & Cronan 2012 — *Remarkable diversity in the enzymes catalyzing the last step in synthesis of the pimelate moiety...* [PMID: 23152908](#). BioG/BioK non-orthologous replacements (F003).

Front-end variation and bypass routes - *A Biotin Biosynthesis Gene Restricted to Helicobacter* (BioV). [PMID: 26868423](#) (F003). - *Three enigmatic BioH isoenzymes... mycobacterial biotin synthesis*. [PMID: 35816546](#) (F003). - *Biochemical and structural characterization of BioZ*. [PMID: 33824341](#). Glutaryl-CoA bypass (F003). - *Pimelic acid / BioW–BioI route in B. subtilis*. [PMID: 28196402](#); review [PMID: 21437340](#) (F003).

Back-end enzymes - *M. smegmatis* BioF (KAPA synthase) structure/function. [PMID: 25462832](#). - Van Arsdell et al. 2005 — *BioA amino-donor variation (SAM vs. lysine)*. [PMID: 15880481](#) (F004). - Cobessi et al. 2012 — *Arabidopsis bifunctional BioA-BioD fusion with channeling*. [PMID: 22547782](#) (F006). - Salaemae et al. 2015 — *M. tuberculosis* DTBS nucleotide promiscuity. [PMID: 25801336](#); ADP binding [PMID: 30289406](#) (F006).

BioB (biotin synthase) - Cramer & Jarrett 2018 — purification/assay of BioB; defines the sulfur-insertion reaction. [PMID: 30097099](#) (F004). - Jameson et al. 2004 — roles of the [2Fe-2S] and [4Fe-4S] clusters. [PMID: 14967042](#) (F004, F007). - Tse Sum Bui et al. 2006 — selenobiotin; [2Fe-2S] as sulfur donor. [PMID: 16533066](#) (F007). - Lachowicz et al. 2024 — auxiliary-cluster diversity (4Fe-5S). [PMID: 38215281](#) (F004, F007). - SAM-[4Fe-4S] interaction. [PMID: 12440894](#).

Boundaries: transport and regulation - Satiaputra et al. 2019 — BirA dual ligase/repressor; alternative DNA-binding. [PMID: 30808984](#) (F008). - Cronan 2024 — BirA specificity/ biotinylation. [PMID: 37997490](#) (F008). - ECF BioMNY transporter mechanism. [PMID: 25991724](#) (F008). - Dethiobiotin salvage via bioYB. [PMID: 37640259](#) (F008). - Alternative repressors BioR / BioX. [PMID: 26037461](#); [PMID: 33972354](#) (F008).

Appendix A: Key Findings and Statistical Evidence (expanded)

F001 — FAS II hijack via methylation/demethylation. In vivo and in vitro reconstitution in *E. coli* showed BioC methylates the free ω -carboxyl of a malonyl-thioester, producing a primer accepted by FAS II; two elongation cycles give pimeloyl-ACP methyl ester, which BioH hydrolyzes to pimeloyl-ACP + methanol. This resolved the long-standing enigma that only *bioC* and *bioH* were required for pimelate synthesis ([PMID: 20693992](#)).

F002 — BioH is the gatekeeper. A 2.05-Å co-crystal structure of *E. coli* BioH bound to pimeloyl-ACP methyl ester demonstrated BioH (not BioF) halts elongation at C7; specificity is enforced through ACP recognition despite BioH being a nonspecific esterase in isolation ([PMID: 23045647](#)).

F003 — Front-end plasticity. *bioH* is absent from many *bioC*-containing genomes and replaced by BioG, BioK, BioV, or Tam-associated activities, or bypassed by BioZ (glutaryl-CoA) or BioW/BioI (free pimelate) ([PMID: 23152908](#); [PMID: 26868423](#); [PMID: 35816546](#); [PMID: 33824341](#)).

F004 — Conserved, ordered ring assembly ending in radical-SAM BioB. BioF condenses pimeloyl-thioester with L-alanine to KAPA; BioA transaminates KAPA to DAPA; BioD closes the ureido ring (ATP-dependent carbamate) to dethiobiotin; BioB inserts sulfur between C6 and C9 using a sacrificial auxiliary cluster ([PMID: 30097099](#); [PMID: 38215281](#)).

F005 — BioC is a malonyl-ACP-specific SAM O-MTase; toxic on overexpression ([PMID: 22965231](#)).

F006 — Eukaryotic BioA-BioD fusion with channeling and mitochondrial targeting ([PMID: 22547782](#); [PMID: 25801336](#)).

F007 — [2Fe-2S] as sacrificial sulfur donor; BioB effectively single-turnover; mechanism still debated ([PMID: 16533066](#); [PMID: 14967042](#); [PMID: 38215281](#)).

F008 — Biosynthesis is bounded by transport and BirA regulation; dethiobiotin salvage overlaps the terminal BioB step ([PMID: 30808984](#); [PMID: 37640259](#)).

Appendix B: Limitations, Knowledge Gaps, and Proposed Follow-up

Limitations and knowledge gaps

1. **Organism heterogeneity.** The canonical model is assembled from *E. coli*, *B. cereus*, *B. subtilis*, mycobacteria, and *Arabidopsis*. Lineage-specific features (BioA donor, BioD nucleotide selectivity, BioB cluster coordination, front-end enzyme identity) mean no single organism represents "the" pathway.
2. **Recalcitrant *E. coli* BioC.** The difficulty of purifying active *E. coli* BioC forces reliance on orthologs, leaving *E. coli*-specific kinetics and regulation incompletely defined.
3. **BioB proximal sulfur species unknown.** Whether the [2Fe-2S] cluster is the immediate donor or generates a persulfide/polysulfide intermediate is unresolved, as is the in vivo cluster-regeneration route that would allow multiple turnovers.
4. **Front-end census incomplete.** The full catalogue of BioH replacements/bypasses across bacterial diversity is still being filled in; new variants continue to be reported.
5. **Structural gaps in ACP-presented steps.** Beyond the BioH complex, high-resolution structures of the FAS-II enzymes acting on the methyl-esterified ACP intermediate are limited.

Proposed follow-up experiments/actions

1. **Trap the BioB sulfur intermediate.** Combine rapid-freeze-quench EPR/Mössbauer with $^{34}\text{S}/^{77}\text{Se}$ labeling and time-resolved mass spectrometry to distinguish direct cluster transfer from a persulfide relay, and identify the residue(s) staging the transferred sulfur.

2. **Reconstitute multi-turnover BioB.** Couple BioB to a defined Fe-S biogenesis/regeneration system in vitro to test whether cluster reloading restores catalytic (>1 equivalent) turnover, quantifying the true throughput ceiling.
3. **Systematic front-end complementation screen.** Use *E. coli* Δ bioH as a reporter to functionally test candidate esterases/bypass enzymes across diverse genomes, building a phylogenetically resolved map of front-end solutions.
4. **Solve active *E. coli* BioC.** Pursue fusion tags, chaperone co-expression, or ortholog chimeras to obtain active, crystallizable *E. coli* BioC for structure and kinetics directly comparable to in vivo data.
5. **Probe the eukaryotic channeling constraint.** Use the *Arabidopsis* BIO3-BIO1 fusion to test whether disrupting the inter-domain channel (linker mutations) reduces flux, quantifying the physiological benefit of channeling versus free diffusion.
6. **Clarify the salvage/synthesis boundary.** Survey *bioY–bioB* organisms for dethiobiotin uptake/conversion and determine how widespread terminal-step-only "synthesis" is across lineages.

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